

Altered Inducibility in KRAS-Independent Pancreatic Cancer: From EMT-Factor Perturbation to DNMT1 and PI3K Hypotheses

An iSURE Research Summary

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) can escape dependence on oncogenic KRAS while retaining an altered response to its inducible expression system. This iSURE project investigated parental AK38 cells and the previously established KRAS-independent recurrent derivative 772. Earlier laboratory experiments showed that knockout of epithelial–mesenchymal transition (EMT)-related transcription factors enhanced doxycycline (Dox)-induced KRAS protein responses without an evident mesenchymal-to-epithelial transition. I integrated transcriptomic comparisons, genotyping, non-targeting controls, immunoblotting and reverse-phase protein array (RPPA) analysis to investigate this phenotype. EMT-associated transcription recurred across withdrawal and relapse models, while 772 recovered proliferative programs and retained a transcriptional Dox response. Genotyping supported a shared model background, and the non-targeting control did not reproduce the knockout-associated response enhancement. Residual MAPK signals, elevated DNMT1 and selective PDK1/AKT-substrate changes identified two priorities: regulation of the inducible system and signaling support for escape growth. These findings provide a framework for testing DNMT1-associated cassette regulation and PI3K-related growth dependence. Methylation measurements and functional perturbations remain necessary to establish causality.

Keywords: PDAC; KRAS independence; EMT; Dox response; DNMT1; PI3K

1 Background and Objectives

Pancreatic ductal adenocarcinoma (PDAC) is an aggressive epithelial malignancy characterized by ductal differentiation and frequent activating *KRAS* mutations. *KRAS* is a small GTPase that regulates growth and survival through pathways including RAF–MEK–ERK and PI3K–AKT–mTOR. Ying et al. established a Dox-inducible *KRAS*^{G12D} mouse model and demonstrated its role in tumor maintenance and anabolic glucose metabolism[2]. Withdrawal induces regression, yet surviving or recurrent populations can emerge. Viale et al. identified mitochondrial dependence in surviving cells, and Kapoor et al. established YAP1/TEAD2-mediated bypass in a subset of recurrent tumors[3, 4].

Using this inducible-model framework, Duong established a recurrent derivative that grew without Dox in culture and after orthotopic reimplantation[1]. This project used the established AK38 parental and 772 recurrent system. *KRAS* independence here denotes growth without inducible oncogenic *KRAS*; endogenous RAS input remains a separate question. Earlier analyses associated recurrence with epithelial–mesenchymal transition (EMT). Subsequent laboratory experiments generated 772 sublines targeting PRRX1, TWIST1 or SNAI1 (Snail1). The sublines showed stronger

Dox-induced KRAS/MAPK protein responses without clear mesenchymal-to-epithelial transition (MET).

My summer work investigated this discrepancy through model-background checks, non-targeting controls and integrated signaling analyses. The central questions were how EMT-factor perturbation affects inducibility and which pathways support continued growth without inducible KRAS (Fig. 1).

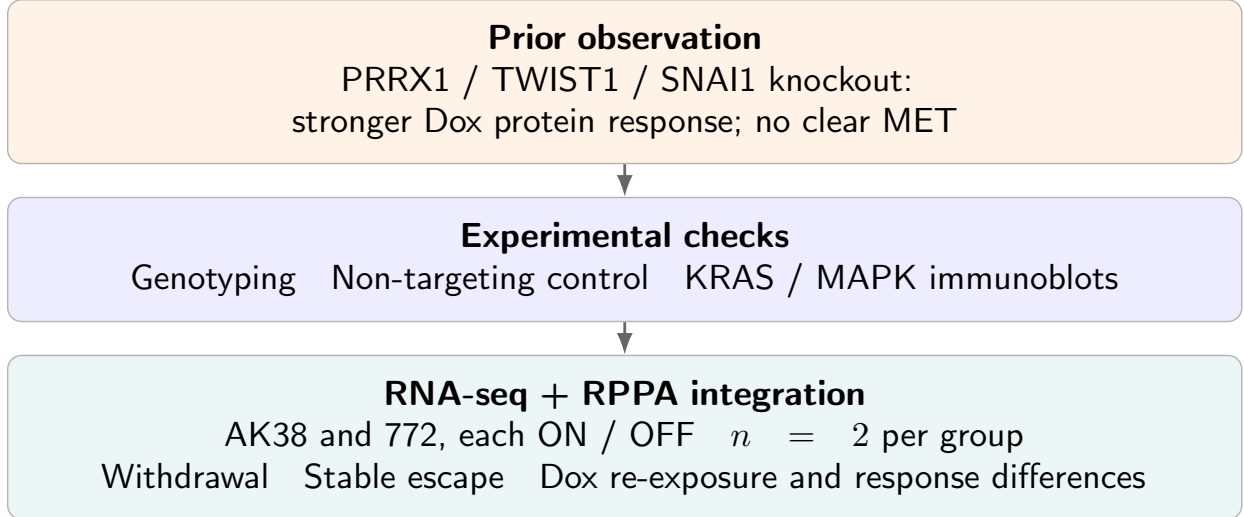


Figure 1: Study rationale and experimental comparisons. Earlier EMT-factor knockout phenotypes motivated the iSURE controls and analyses. The four-group omics cohort examines withdrawal, escape and Dox re-exposure; knockout sublines and non-targeting controls were assessed separately. Arrows indicate the investigative sequence; each omics group contains two biological replicates.

2 Study Design

RNA-seq and RPPA included AK38 OFF, AK38 ON, 772 OFF and 772 ON, with two biological replicates per condition. ON/OFF denote Dox presence/absence. AK38 OFF versus ON measures withdrawal; 772 OFF versus AK38 OFF measures escape relative to a matched Dox-free control; 772 ON versus OFF measures retained responsiveness. The interaction compares the two within-line Dox responses. Genotyping and immunoblotting assessed AK38, 772 and 772_sgCon. Detailed protocols and statistical thresholds appear in the Supplementary Methods.

Three public mouse PDAC datasets provided complementary comparisons. GSE32277 compares 24-hour Dox withdrawal with maintained induction in cultured cells and orthotopic tumors, testing whether EMT accompanies early withdrawal[2]. GSE58307 compares surviving cells (SCs) with KRAS-expressing spheres from matched tumors, distinguishing residual survival from stable proliferative escape[3]. GSE53169 compares KRAS-independent or KRAS-reexpressing relapse with parental ON, testing reproducibility and specificity of the relapse-associated EMT program[4].

3 Results

3.1 772 combines EMT-associated transcription with proliferative recovery and retained responsiveness

AK38 withdrawal enriched EMT (NES 2.512, $q = 8.14 \times 10^{-14}$) while reducing E2F/G2M, MYC and MAPK-related output. Relative to AK38 OFF, 772 OFF retained EMT enrichment (NES 2.003, $q = 1.92 \times 10^{-6}$) and recovered proliferative programs (Fig. 2A). Glycolysis increased relative to AK38 OFF but not significantly relative to AK38 ON, illustrating the importance of the reference condition.

Dox re-exposure increased MAPK, PI3K and proliferative transcription in unmodified 772 (Fig. 2B). Thus, the original line retained transcriptional responsiveness, whereas the knockout-associated observation concerned stronger KRAS protein and downstream responses. The negative MAPK interaction describes a smaller response than in AK38 and is compatible with a positive response within 772.

Public analyses supported EMT enrichment after withdrawal in GSE32277 and in both relapse classes in GSE53169 (Supplementary Figs. S3A and S7B). The direct relapse-class comparison provides additional context (Supplementary Fig. S3B). GSE58307 distinguished mitochondrial, fatty-acid oxidation and lysosomal programs in SCs from E2F/G2M and glycolytic recovery in 772 (Supplementary Fig. S7A). These comparisons support recurring plasticity across distinct states; their functional and lineage relationships remain to be tested.

A

	Withdrawal AK38 OFF / ON	Stable escape T72 OFF / AK38 OFF	Induced reference T72 OFF / AK38 ON	Re-exposure T72 ON / OFF
EMT	+2.51*	+2.00*	+2.65*	+1.22
E2F targets	-2.14*	+3.07*	+2.38*	+2.08*
G2M checkpoint	-2.11*	+2.55*	+1.68*	+2.25*
MYC targets V1	-1.83*	+2.44*	+1.83*	+2.13*
Glycolysis	-1.84*	+1.58*	-1.13	+1.25
mTORC1 signaling	-2.02*	+1.79*	-1.15	+1.84*
KRAS signaling up	-2.01*	-1.21	-1.92*	+1.69*
TGF- β signaling	+1.90*	-2.00*	-1.30	+1.46
Oxidative phosphorylation	-1.08	-1.15	-1.31	+1.05

Signed NES: blue = negative; red = positive; white = zero.

* $q < 0.05$ within the original 50-Hallmark analysis; no new significance testing.

B

	Withdrawal AK38 OFF / ON	Stable escape T72 OFF / AK38 OFF	Induced reference T72 OFF / AK38 ON	Re-exposure T72 ON / OFF	Response difference Interaction
MAPK	-11.39*	+7.55*	-1.28	+10.77*	-5.74*
PI3K	+0.01	-0.46	-0.30	+9.27*	+4.53*
TGF- β	+10.10*	-0.24	+5.93*	+1.12	+10.51*

Signed model statistic; blue = negative; red = positive.

* $q < 0.05$ using the original pathway correction within each contrast.

Figure 2: Transcriptional plasticity, proliferative recovery and Dox responsiveness. (A) Nine Hallmark programs across four contrasts; stars indicate $q < 0.05$ within each original 50-Hallmark analysis. Positive NES indicates enrichment toward the first group. (B) Three selected PROGENy top-500 multivariate-model statistics; stars indicate $q < 0.05$ using the original pathway correction within each contrast. Interaction scores describe differences between within-line responses. Both panels use two biological replicates per condition and report transcriptional associations.

3.2 Model controls support a perturbation-associated change in protein inducibility

AK38 and 772 showed concordant tested genotyping markers, supporting their shared tumor-model origin together with the established derivation and tumor-forming behavior[1]. The 772_sgCon control retained the original protein-response phenotype and did not reproduce the knockout-associated enhancement. These findings focus attention on changes associated with the targeted perturbations (Supplementary Table S1; Figs. S8, S6A–B and S6C–D).

Immunoblots showed total MEK/ERK and residual phospho-MEK/phospho-ERK bands in examined Dox-free 772 panels. Their upstream source requires direct testing. RPPA showed no significant p-MEK difference between 772 OFF and AK38 OFF; increased p-ERK was measured with a use-with-caution antibody (Supplementary Fig. S4). These protein observations complement the retained transcriptional response described above.

3.3 DNMT1 and selective phosphoprotein changes define two mechanistic priorities

DNMT1 was elevated in 772 OFF relative to AK38 OFF (log-scale effect +1.271, $q = 0.00336$) and AK38 ON (effect +1.007, $q = 0.00969$; Fig. 3A–B). Duong previously reported elevated DNMT1 and increased sensitivity to GSK-3484862[1]. The present question is whether DNMT1-associated methylation regulates the inducible construct and contributes to the response changes after EMT-factor perturbation.

RPPA also identified increased PDK1 and selected AKT-substrate-associated paired signals, including GSK3 and PRAS40 (Fig. 3C). PRAS40 includes a use-with-caution total-protein antibody. Neither direct AKT phosphosite was significant; mTOR was suggestive, and S6K, S6 and 4E-BP1 lacked coordinated enhancement. In the matched-OFF comparison, no Reactome set reached $q \leq 0.05$. The matched-OFF PI3K transcriptional footprint was also nonsignificant ($q = 0.819$). These selective changes prioritize functional testing of growth support alongside the separate question of inducible-system regulation.

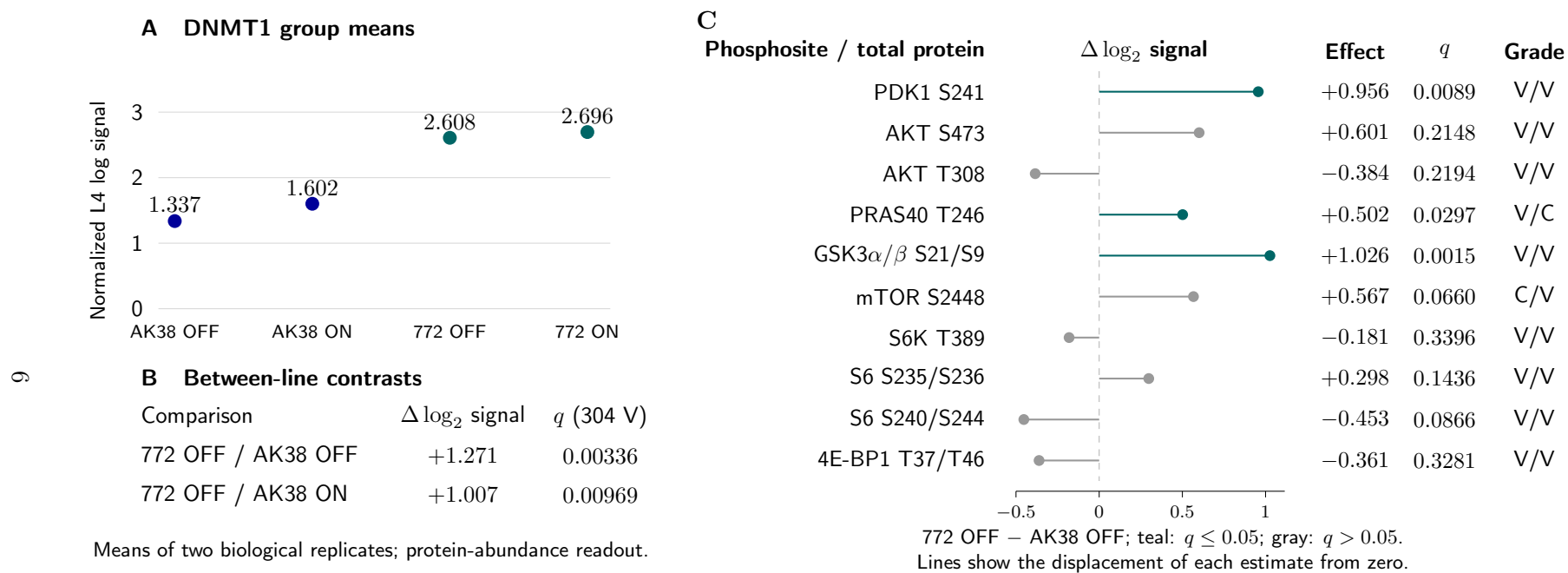


Figure 3: Protein evidence prioritizes inducible-system regulation and escape-growth support. (A) DNMT1 normalized L4 means, each summarizing two biological replicates. (B) Four-group limma contrasts, BH-adjusted across 304 validated antibodies. (C) Selected phosphosite-minus-total-protein effects for 772 OFF versus AK38 OFF, BH-adjusted across 53 pairs. Lines show displacement from zero; teal indicates $q \leq 0.05$. V/C denote validated/use-with-caution antibodies in phosphosite/total order. Abundance and paired-signal effects motivate hypotheses; methylation and functional dependence require direct testing.

4 Discussion and Next Steps

The motivating observation links EMT-factor perturbation to enhanced protein inducibility with no clear MET. It therefore calls for separate assessment of cellular state, induction-system regulation and growth dependence. Viale’s surviving cells also retained responses to KRAS re-expression[3]. Kapoor’s YAP bypass demonstrates an alternative route to proliferative output[4]; its contribution in 772 requires targeted validation.

Two experiments take priority. First, compare unmodified 772, 772_sgCon and independent knockout sublines under matched Dox conditions, combining DNMT1 perturbation with construct-region methylation, KRAS RNA/protein and EMT measurements. An orthogonal reporter or Dox-independent expression control would help localize the regulatory change. Second, test PI3K inhibition in 772 and AK38 ON/OFF states, pairing growth measurements with target engagement and independent pharmacological or genetic validation. MEK-pathway controls would distinguish selective dependencies from shared growth requirements.

5 Limitations

The omics cohort has two biological replicates per condition and one recurrent derivative. Cross-model enrichment is associative, and cell identity checks cover only the tested markers. Clonal and off-target effects require additional controls. The knockout/MET observation remains preliminary; WB panels lack densitometry, complete membranes and confirmed replication. Cassette methylation, pathway dependence and a lineage connecting residual survival to stable escape remain unresolved.

6 Conclusion

This iSURE project combined transcriptomic analysis, model-background and non-targeting controls, immunoblotting and RPPA to refine the explanation of altered inducibility in 772. The findings prioritize two hypotheses for functional testing: DNMT1-associated regulation of the induction system and PI3K-related support of escape growth. Inducibility, EMT state and KRAS growth dependence remain distinct experimental readouts.

References

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S1 Supplementary Methods

S1.1 Internal cohort, counting and quality control

The mouse KRAS^{G12D}-inducible cohort included AK38 OFF (LDD1–LDD2), AK38 ON (LDD3–LDD4), 772 OFF (LDD5–LDD6) and 772 ON (LDD7–LDD8), with $n = 2$ biological replicates per group. Samples LDD9–LDD18 were not included. RNA sequencing used paired-end 150-bp reads aligned with HISAT2 2.2.1 to NCBI GRCm39 (GCF_000001635.27). Reverse-stranded, paired-end fragment counting used featureCounts 2.0.6 and the matching annotation, excluding multimapping and multi-overlapping fragments[5]. Mapped-read fractions were 98.29–98.90%, and assignment rates were 70.1–77.9%. Requiring at least 10 counts in at least two samples retained 14,486 of 41,079 gene-level features.

S1.2 Differential expression and interaction analysis

Pairwise comparisons used a DESeq2 four-group design[6]. Positive $\log_2$ fold changes indicate higher expression in the first named group. Gene-level significance was defined as Benjamini–Hochberg (BH)-adjusted $P < 0.05$; directional lists additionally required $|\log_2 \text{FC}| \geq 1$. The factorial model, with AK38 and OFF as reference levels, was

$$\log_2 \left(\frac{\mu_{gj}}{s_j} \right) = \beta_{g,0} + \beta_{g,\text{cell}} C_j + \beta_{g,\text{Dox}} D_j + \beta_{g,\text{int}} C_j D_j.$$

Here, μ_{gj} is the expected count for gene g in sample j , and s_j is the sample size factor. The cell-line indicator C_j equals 1 for 772 and 0 for AK38; the Dox indicator D_j equals 1 for ON and 0 for OFF. This is the log-mean relationship of the DESeq2 negative-binomial generalized linear model, with $\beta_{g,0}$ representing the AK38 OFF baseline. For each gene, the interaction coefficient $\beta_{g,\text{int}}$ estimates the difference between the two within-line Dox responses:

$$\Delta_{\text{int}} = \beta_{g,\text{int}} = (772_{\text{ON}} - 772_{\text{OFF}}) - (AK38_{\text{ON}} - AK38_{\text{OFF}}). \quad (1)$$

The group symbols denote the fitted $\log_2$ normalized expected expression of gene g in each condition. A positive interaction indicates a more positive Dox-response effect in 772 than in AK38. Variance-stabilized counts supported principal-component analysis (PCA) and sample-level visualization.

S1.3 Enrichment, signaling footprints and external comparisons

Hallmark GSEA used DESeq2 Wald-statistic rankings[7, 8]. GO/KEGG over-representation analyses used separate upregulated and downregulated lists and a tested, symbol-resolved gene background. PROGENy top-500 footprints and multivariate linear models estimated pathway-level responses; CollecTRI regulons and univariate models estimated transcription-factor (TF) output through decoupleR[9, 10, 11]. Alternative models, DoRothEA regulons, sample-level limma tests and removal of E2F/G2M genes provided sensitivity analyses. Footprint scores are model-derived statistics of pathway responses or transcription-factor output.

Public GSE32277, GSE58307 and GSE53169 data were reanalyzed using dataset-specific RMA normalization and limma[12]. GSE58307 retained its matched-tumor structure; independent lines were the statistical units in GSE32277. GSE53169 separated KRAS-independent, KRAS-reexpressed and YAP1-associated relapse categories. Integration used common-symbol rankings, pathway NES correlations and signature overlap, with RNA-seq and microarray intensities analyzed separately.

Unless otherwise stated, q denotes the BH-adjusted value within the relevant analysis family; each testing family is corrected separately.

S1.4 RPPA and evidence grading

The RPPA Set201 L4 matrix contained 500 antibodies across eight samples, analyzed together in a four-group limma model. Primary individual-protein calls required validated, non-discontinued antibodies, $q \leq 0.05$ within the 304 validated-antibody family and $|\log_2 \text{effect}| \geq 0.5$. Use-with-caution (C-grade) antibodies remained exploratory. Phosphosite-minus-total-protein contrasts were fitted jointly and BH-adjusted across 53 pairs. These contrasts compare separately normalized log-scale signals to describe relative phosphorylation changes. Reactome competitive tests are interpreted within the measured targeted panel.

S2 Additional Analytical Methods

S2.1 Pathway selection and visualization

Nine Hallmark programs were selected to examine cellular plasticity, proliferation, metabolism and KRAS-associated transcription across four contrasts (Fig. 2). Signed NES values are displayed on a common color scale without row scaling or clustering. Significance follows the original correction within each 50-Hallmark comparison. The RPPA display (Fig. 3) presents ten phosphosite/total-protein readouts spanning the candidate signaling axis, with BH correction across all 53 paired readouts. These figures emphasize selected readouts relevant to the research questions. Design and working-model diagrams organize the experimental comparisons and mechanistic hypotheses by summarizing existing observations and links to be tested.

S2.2 DNMT1 abundance analysis

The DNMT1 display uses normalized L4 group means and between-line effects from the four-group RPPA model. DNMT1 is a validated, non-discontinued antibody; its reported q values use BH correction across 304 validated antibodies; correction across all 500 antibodies constitutes a separate testing family. These abundance measurements are distinct from the 53 phosphosite/total-protein paired readouts. DNMT1 methylation activity and methylation of specific construct regions were not measured in this analysis.

S2.3 Cross-dataset comparisons

Public datasets GSE32277, GSE58307 and GSE53169 were reanalyzed to compare withdrawal, surviving-cell and recurrent states[2, 3, 4]. The harmonized 50-Hallmark and focused nine-program GSE58307 analyses differ in their values and correction contexts: SC EMT has NES 1.360, $q = 0.0685$ in the former and NES 1.357, $q = 0.0226$ in the latter. Their significance values are interpreted within the respective testing families. For SC versus sphere and 772 OFF versus AK38 OFF, same-direction top-250 overlaps were zero upregulated and five downregulated genes within a 12,770-gene common background; both Fisher tests had adjusted $q = 1.0$. Receptor modules in Supplementary Fig. S5 can share the same downstream footprint and require independent validation of each receptor contribution.

S2.4 Model-background and non-targeting controls

Genotyping compared AK38 and 772 at the tested model-associated markers. The gel is evaluated for concordance of the labeled cell samples, without assigning unverified allele sizes or interpreting it as complete lineage authentication. The non-targeting line 772_sgCon was compared with the unmodified 772 line under matched Dox conditions to test whether the control manipulation reproduced the knockout-associated protein-response restoration. EMT-factor knockout phenotypes are preliminary laboratory findings and are not part of the four-group RNA-seq design. The stated $n = 2$ omics replication must not be transferred to genotyping, knockout or WB experiments.

S2.5 Immunoblot readouts and image handling

Figures S6A–B and S6C–D show immunoblot results for AK38, 772 and 772_sgCon cells with and without Dox. The measured proteins include KRAS, EMT-associated proteins, MEK/ERK and their phosphorylated forms. Vinculin is included as a loading-control readout. Lane order is preserved within each panel; the two panels in Fig. S6C–D use different Dox orders and should not be compared column-for-column without checking the labels.

Blot strips are presented as cropped images without brightness or contrast adjustment. No densitometry or statistical testing was performed, and image counts are not interpreted as biological replicate counts. Full uncropped membranes, exposure conditions, antibody details, molecular-weight markers and membrane pairing are required for quantitative assessment. The images provide qualitative records of protein expression and phosphorylation; functional dependence requires further experiments. The sgCon comparison tests whether the non-targeting control reproduces the marked response restoration; other CRISPR-associated effects require additional controls.

S3 Supplementary Tables

Table S1: Experimental controls and their implications for the Dox-response phenotype.

Question	Observation	Supported interpretation
Does 772 retain the model background?	AK38 and 772 share the expected tested marker patterns.	Supports a parental-recurrent relationship; complete identity requires further authentication.
Does the control manipulation restore the response?	772_sgCon does not reproduce the marked knockout-associated restoration.	A generic non-targeting manipulation is insufficient in this comparison.
Is response restoration equivalent to MET?	Earlier knockout sublines show stronger inducibility without an evident MET.	Inducible-system responsiveness and epithelial state must be assessed separately.

Table S2: Gene-level differential-expression and interaction results.

Contrast	Tested	$q < 0.05$	Up	Down
AK38 OFF – AK38 ON	14,486	8,540	1,639	1,823
772 OFF – AK38 OFF	14,486	10,422	2,442	2,876
772 OFF – AK38 ON	14,486	10,001	2,077	2,674
772 ON – 772 OFF	14,486	5,504	357	583
772 ON – AK38 ON	14,486	10,410	2,278	3,101
Cell line $\times$ Dox	14,486	7,527	1,669	1,921

Up and Down additionally require $\log_2 \text{FC} \geq 1$ and ≤ -1 , respectively. Interaction directions follow Eq. 1. All models use eight samples, with two biological replicates per condition.

Table S3: EMT enrichment across experimental contrasts and public datasets.

Contrast	NES	q
AK38 OFF – AK38 ON	+2.512	8.14×10^{-14}
772 OFF – AK38 OFF	+2.003	1.92×10^{-6}
772 OFF – AK38 ON	+2.651	5.21×10^{-17}
GSE32277: culture OFF – ON	+2.272	5.58×10^{-13}
GSE32277: tumor OFF – ON	+2.632	1.64×10^{-17}
GSE53169: independent relapse – parental ON	+3.282	9.42×10^{-29}

Internal and public comparisons were analyzed separately using Hallmark GSEA. Positive NES indicates enrichment toward the first group. Cross-dataset NES values compare transcriptional enrichment relative to the respective reference conditions.

Table S4: Selected phosphosite-minus-total RPPA effects: 772 OFF versus AK38 OFF.

Readout	$\Delta \log_2 \text{ signal}$	q	Grade
p-PDK1 S241 / PDK1	+0.956	0.0089	V/V
p-AKT S473 / AKT	+0.601	0.2148	V/V
p-AKT T308 / AKT	-0.384	0.2194	V/V
p-PRAS40 T246 / PRAS40	+0.502	0.0297	V/C
p-GSK3 α/β S21/S9 / GSK3	+1.026	0.0015	V/V
p-mTOR S2448 / mTOR	+0.567	0.0660	C/V
p-S6K T389 / S6K	-0.181	0.3396	V/V
p-S6 S235/S236 / S6	+0.298	0.1436	V/V
p-S6 S240/S244 / S6	-0.453	0.0866	V/V
p-4E-BP1 T37/T46 / 4E-BP1	-0.361	0.3281	V/V

The slash denotes a phosphosite/total pair, analyzed as a difference of normalized log signals, describing relative phosphorylation changes. Grade lists phosphosite/total antibody status: V, validated; C, use with caution. BH correction covers 53 pairs; each condition has two biological replicates.

S4 Supplementary Figures

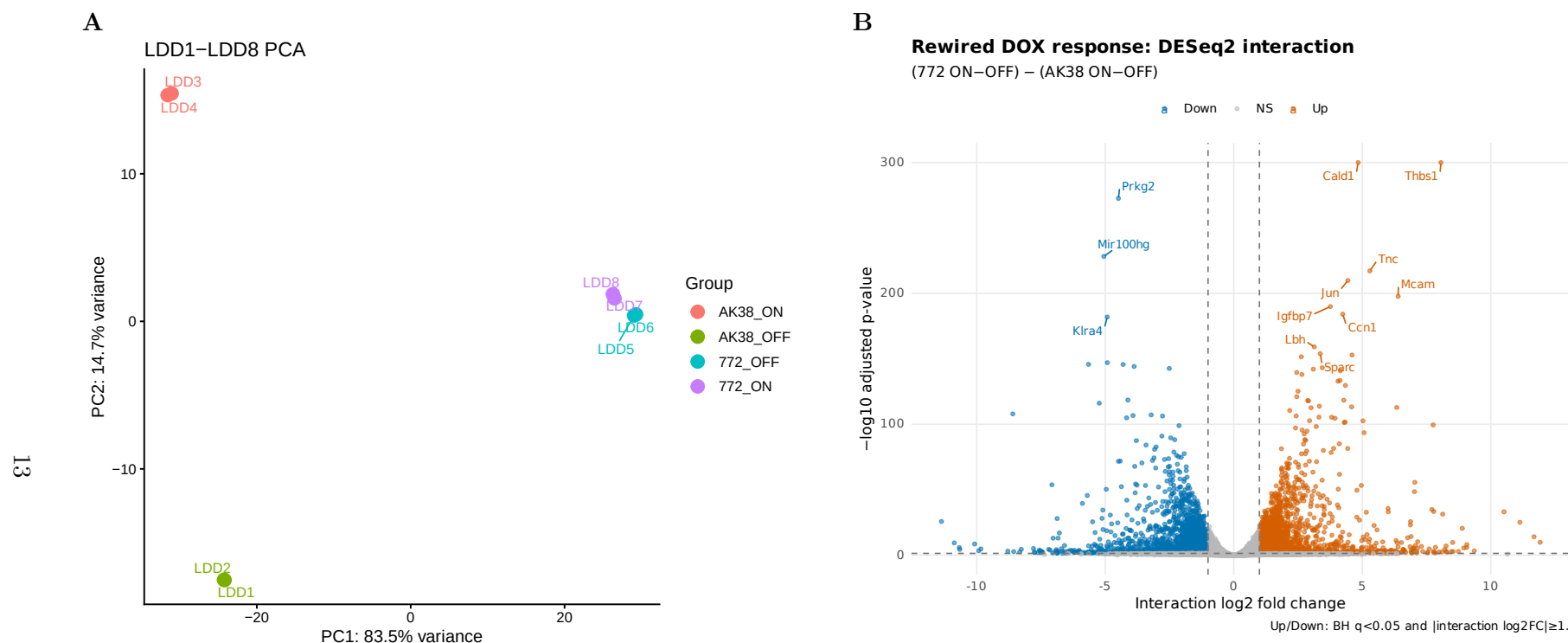
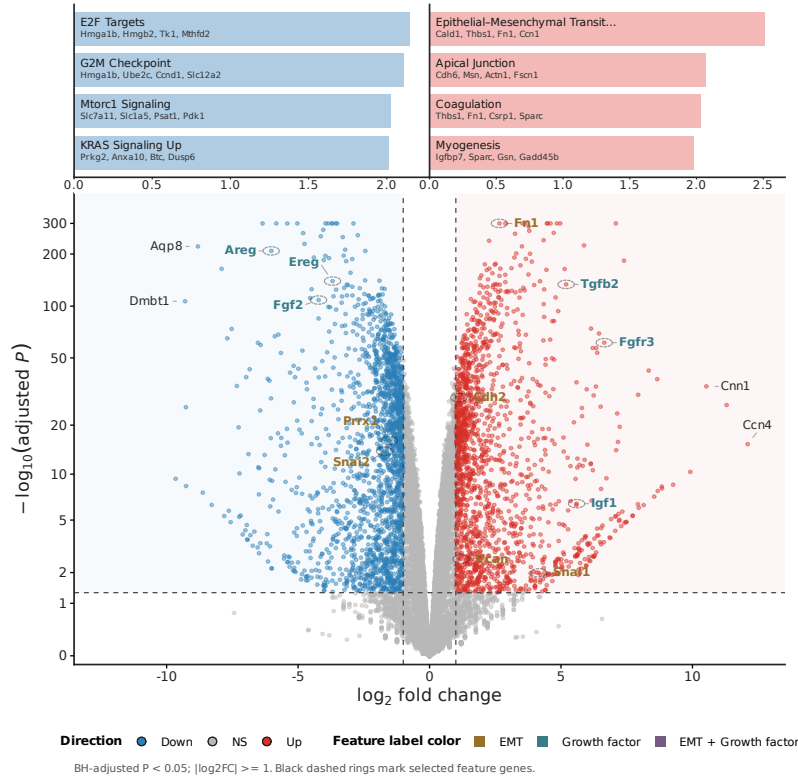


Figure S1: Cohort structure and differential Dox responses. (A) PCA of variance-stabilized counts for 14,486 retained genes. LDD1–2 are AK38 OFF, LDD3–4 AK38 ON, LDD5–6 772 OFF and LDD7–8 772 ON; each point is one biological replicate. PC1/PC2 explain 83.5%/14.7% of variance. (B) DESeq2 interaction volcano corresponding to Eq. 1. Positive effects indicate a more positive ON-versus-OFF response in 772 than in AK38. Each condition has two biological replicates; gene-level thresholds are specified in the Supplementary Methods.

A

AK38 OFF vs ON

Differential expression volcano plot with Hallmark GSEA enrichment (independent absolute NES axes)



B

772 OFF vs AK38 OFF

Differential expression volcano plot with Hallmark GSEA enrichment (independent absolute NES axes)

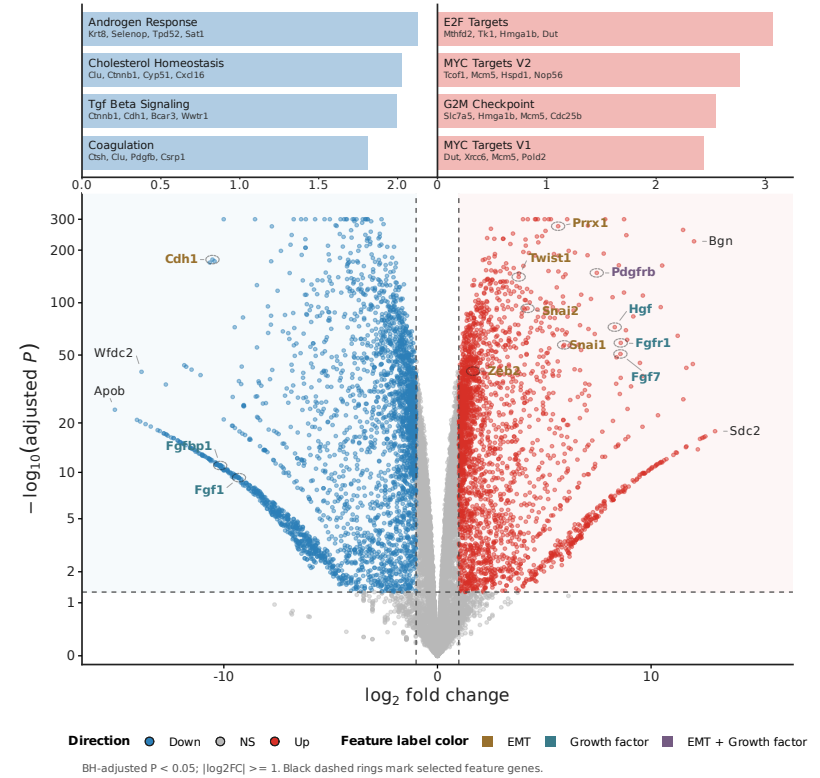
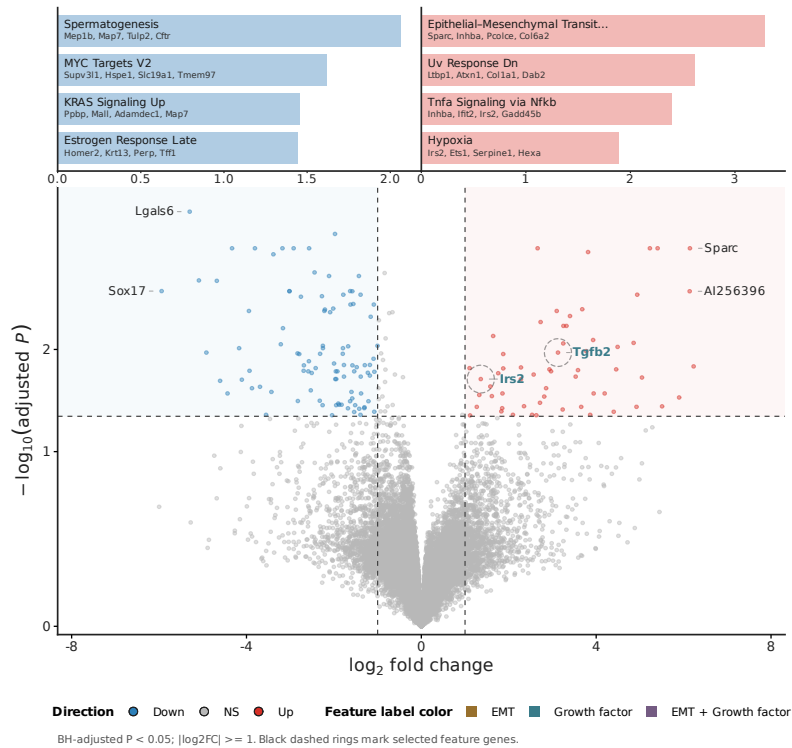


Figure S2: Internal KRAS withdrawal and stable escape. (A) AK38 OFF versus AK38 ON. (B) 772 OFF versus AK38 OFF. Each group contains $n = 2$ biological replicates; positive effects indicate higher expression in the first group. Colored points satisfy BH-adjusted $P < 0.05$ and $|\log_2 FC| \geq 1$; dashed rings highlight selected genes. Upper panels show selected negative and positive Hallmark enrichments on independent absolute-NES axes. Panel B highlights E2F/G2M/MYC recovery; EMT values are given in Table S3.

A

GSE53169: KRAS-independent relapse (E1-E8) vs parental KRAS-ON

Differential expression volcano plot with Hallmark GSEA enrichment (independent absolute NES axes)



B

GSE53169 relapse: KRAS-independent (E1-E8) vs KRAS-reexpressed (E9-E16)

Differential expression volcano plot with Hallmark GSEA enrichment (independent absolute NES axes)

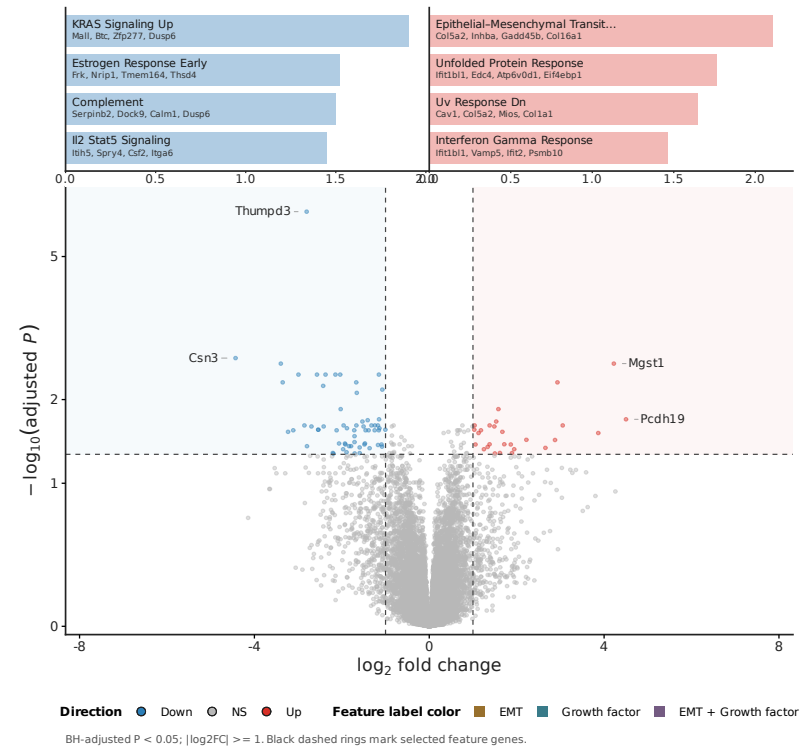


Figure S3: Public comparisons of KRAS-independent relapse. RMA/limma reanalysis of GSE53169[4]. (A) KRAS-independent E1–E8 relapse lines versus two parental KRAS-ON lines. (B) E1–E8 versus KRAS-reexpressing E9–E16 relapse lines. Positive effects indicate higher expression in independent relapse. Colored points satisfy BH-adjusted $P < 0.05$ and $|\log_2 FC| \geq 1$; dashed rings highlight selected genes. Upper panels use independent absolute-NES axes for negative and positive Hallmark enrichment. The comparisons assess recurrent EMT-associated transcription and differences between relapse states.

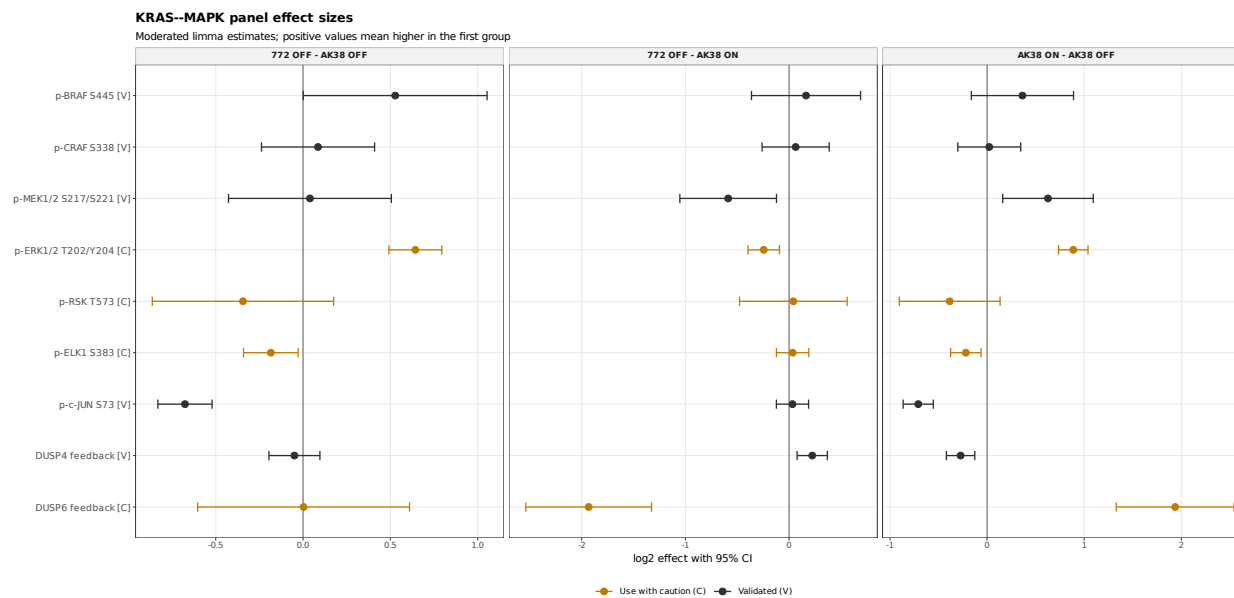


Figure S4: MAPK RPPA contrast estimates. Points and intervals show fitted log-scale effects and 95% confidence intervals across the displayed contrasts. All eight samples entered a four-group limma model ($n = 2$ per condition). V-grade effects use the validated-antibody correction; C-grade p-ERK is exploratory. Relative to AK38 OFF, p-ERK increased and p-MEK showed no significant difference.

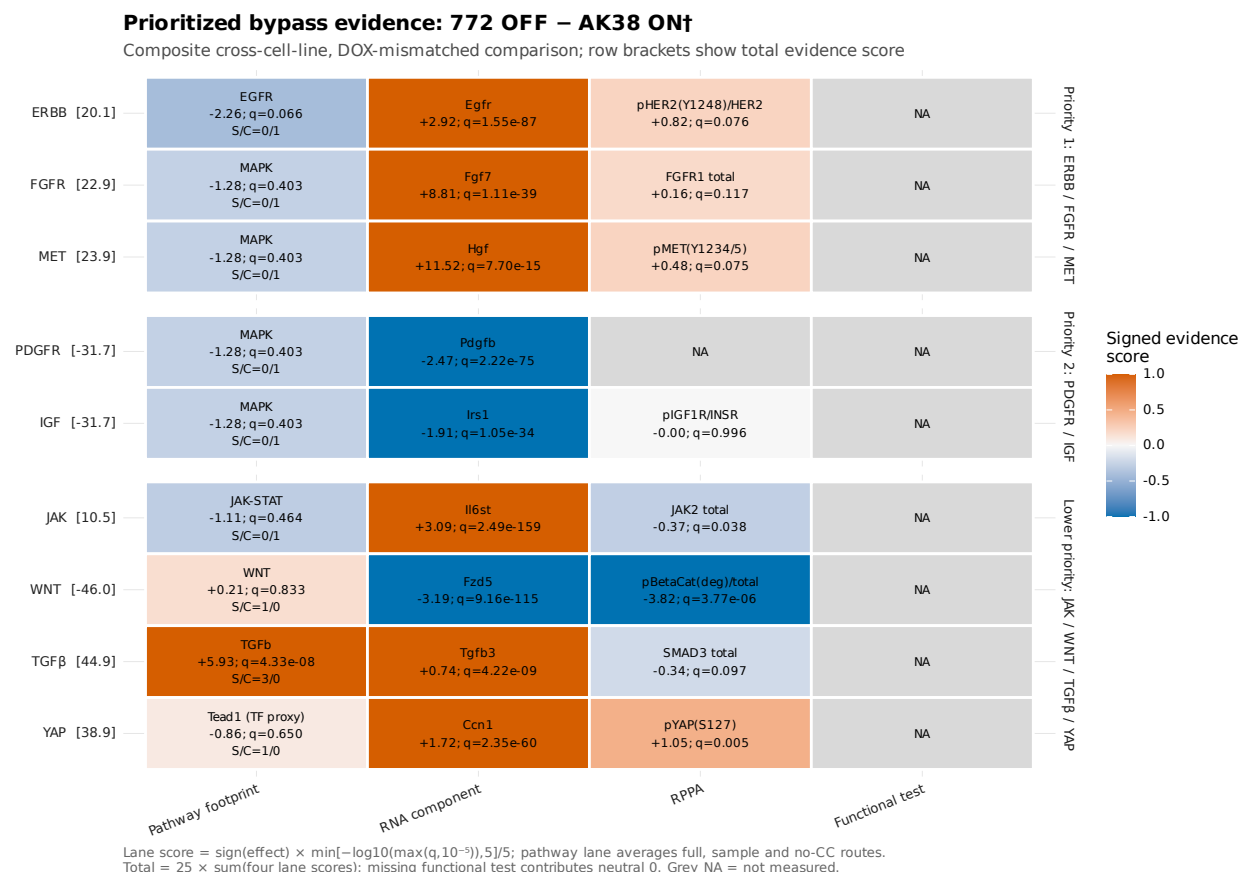
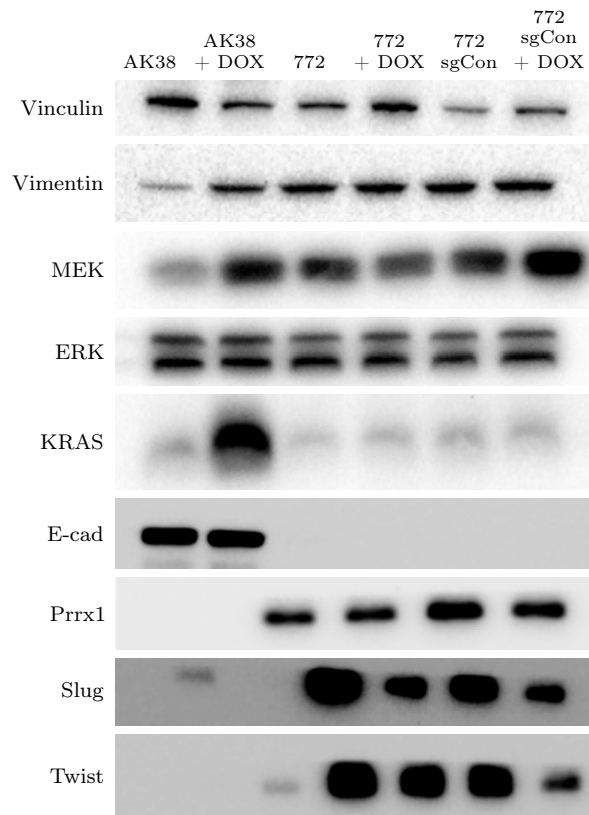
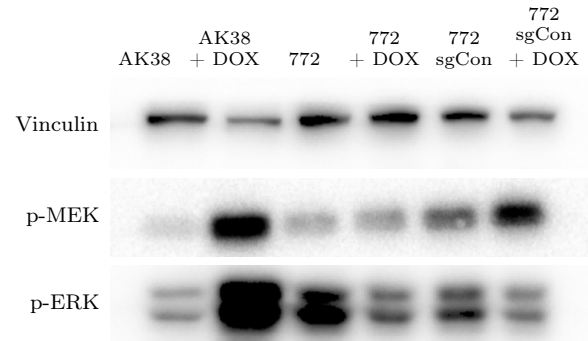


Figure S5: Candidate bypass evidence matrix. The figure assembles analytical evidence for candidate prioritization. Receptor modules share MAPK footprints, and receptor-specific contributions require further testing; missing entries indicate unavailable evidence. The model-dependent and incomplete evidence motivates functional testing.

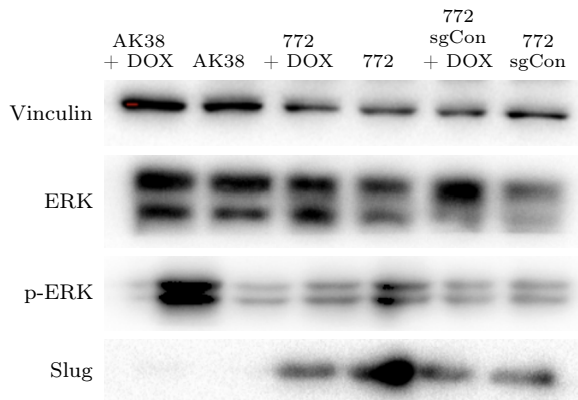
A Protein / EMT panel



B Phosphoprotein panel



C 0820



D 0806

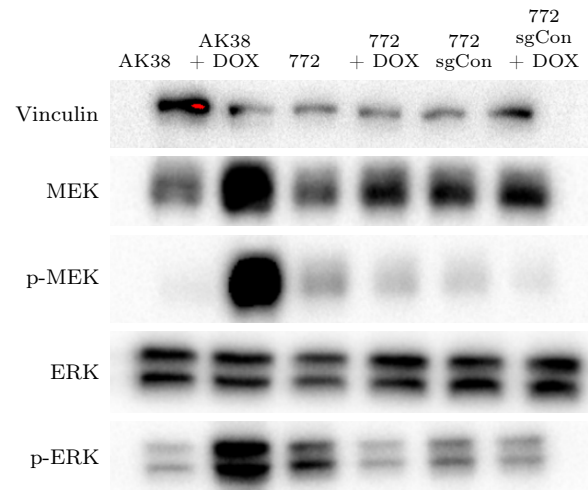
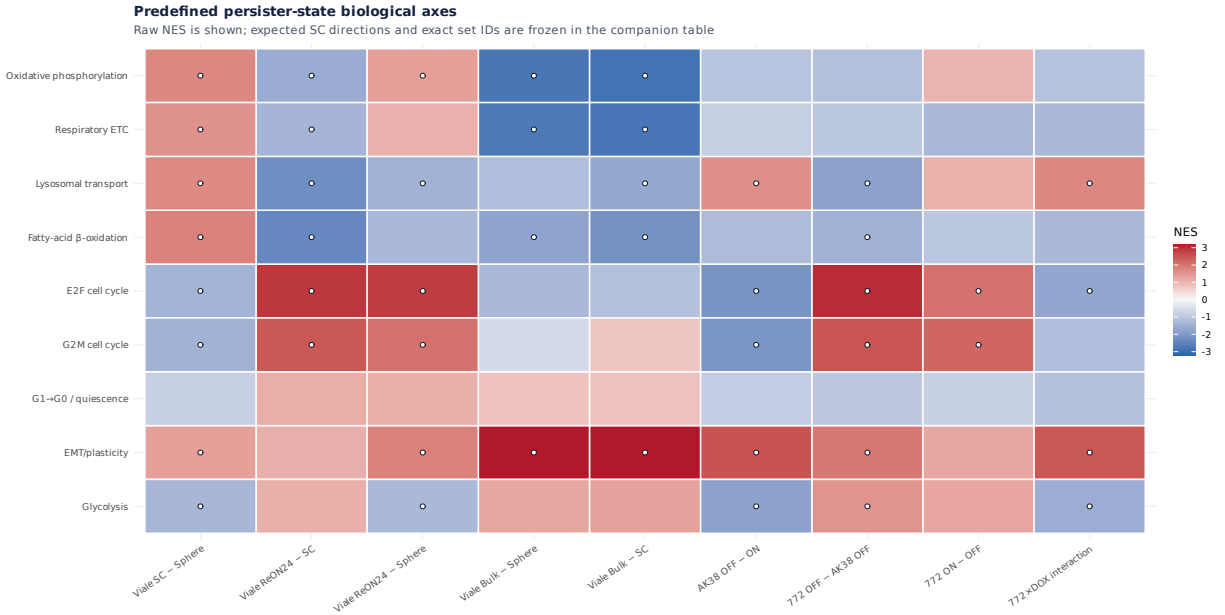


Figure S6: Protein expression and MAPK phosphorylation across AK38, 772 and the non-targeting control. (A) Vinculin, Vimentin, MEK, ERK, KRAS, E-cad, Prrx1, Slug and Twist. (B) Vinculin, p-MEK and p-ERK. (C) Panel 0820: Vinculin, ERK, p-ERK and Slug. (D) Panel 0806: Vinculin, MEK, p-MEK, ERK and p-ERK. In A, B and D, lanes are AK38, AK38 + DOX, 772, 772 + DOX, 772_sgCon and 772_sgCon + DOX. Panel C reverses the ON/OFF order within each cell line. Display crops and lane order are preserved; brightness, contrast and intensities are unchanged. These are qualitative records without densitometric normalization. Panel labels 0820/0806 do not specify a year or independent replicate count.

A



B

	GSE32277: culture OFF / ON	GSE32277: tumor OFF / ON	GSE58307: SC / sphere Surviving state	GSE53169: independent relapse / parental ON	GSE53169: reexpressing relapse / parental ON
EMT	+2.27*	+2.63*	+1.36	+3.28*	+3.19*
E2F targets	-1.85*	-2.76*	-1.40*	-1.17	-2.02*
G2M checkpoint	-1.36*	-2.20*	-1.45*	-0.84	-1.29
MYC targets V1	-2.50*	-2.63*	-1.57*	-1.31	-2.09*
Glycolysis	-1.15	-1.36*	-1.35	-0.82	-0.86
mTORC1 signaling	-2.67*	-2.66*	-2.31*	-1.13	-1.73*
KRAS signaling up	-1.90*	-1.95*	-1.89*	-1.45*	+1.37*
TGF-beta signaling	+2.13*	+1.31	+0.97	+1.88*	+2.18*
Oxidative phosphorylation	+1.74*	-1.20	+1.75*	-1.00	-1.58*

Signed NES: blue = negative; red = positive; white = zero.
* $q < 0.05$ within each original 50-Hallmark analysis; common scale across contrasts.

Figure S7: Public programs distinguish residual survival and relapse. (A) Nine predefined programs across the indicated GSE58307 and internal contrasts. GSE58307 contains five samples per state, including SCs and matched KRAS-expressing spheres; internal conditions contain two biological replicates[3]. Colors and dots indicate signed NES and significance. SC EMT has $q = 0.0226$ in this focused analysis. (B) Nine selected Hallmark programs from the original 50-Hallmark analyses: GSE32277 culture/tumor OFF versus ON; GSE58307 SC versus sphere; and GSE53169 independent/reexpressing relapse versus parental ON[2, 3, 4]. Stars indicate $q < 0.05$ in each original testing family; SC EMT has $q = 0.0685$ here. Both panels describe transcriptional states. Scores retain their original normalization and significance, with no new tests.

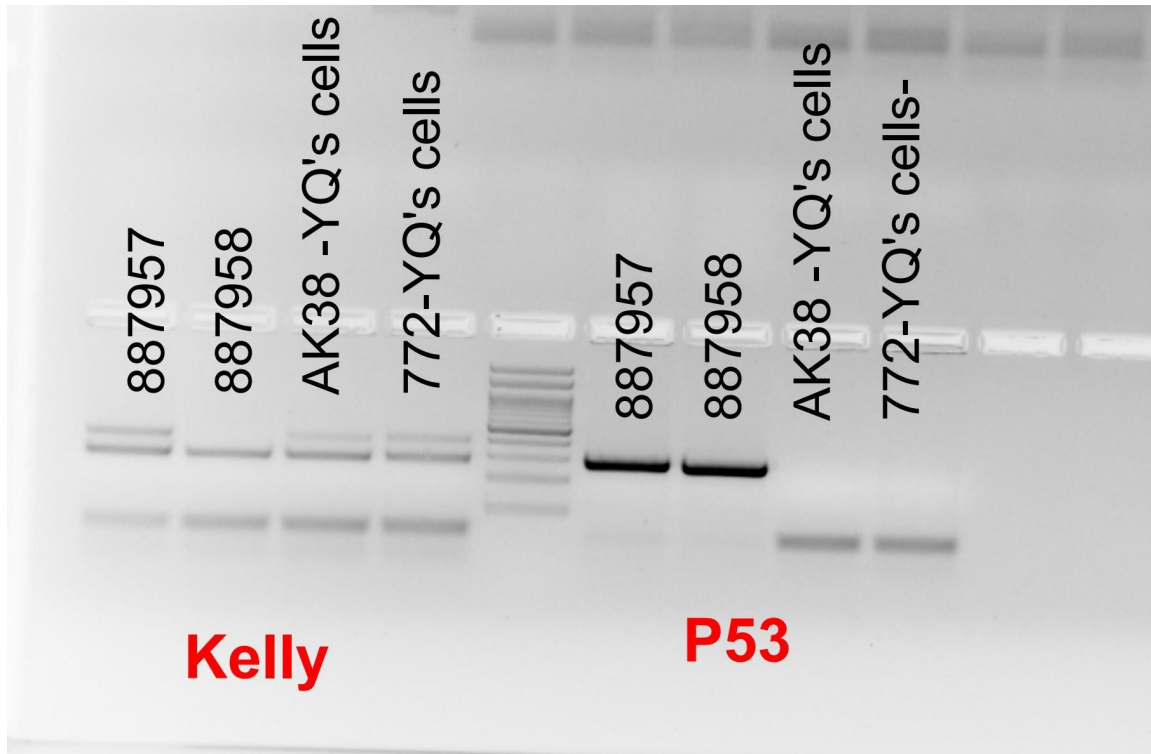


Figure S8: Genotyping supports concordance at the tested model markers. The gel retains its sample order and labels, including AK38, 772, 887957 and 887958, and the assay labels Kelly and P53. The two labeled AK38/772 comparisons show concordant patterns within each assay. No additional allele sizes, identities for the numbered samples, or molecular target for the Kelly assay are assigned. This qualitative comparison supports concordance at the tested markers; complete identity, population purity and cassette structure require further authentication. The image is unchanged, and independent replicate numbers are not inferred.